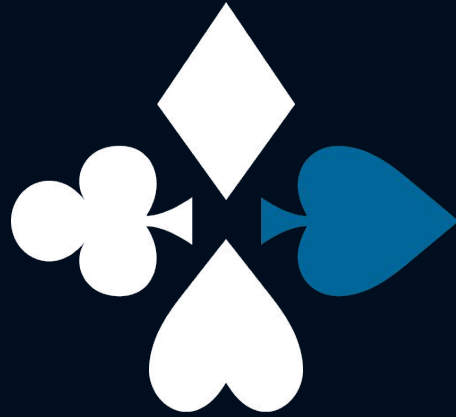




Pokerbots 2026

Lecture 2: Probability and Statistics

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Register & Resume Drop:
pkr.bot/reg

No Prize Giveaway!

pkr.bot/clubs

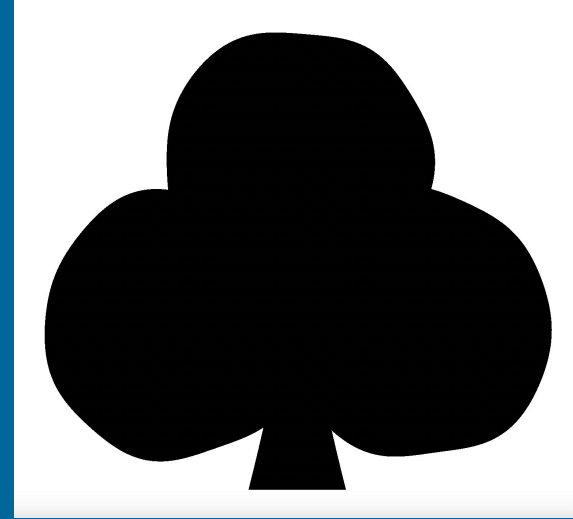


Guess the percentage of pixels in this chunky club image which are not white (resolution 382x350)

Feel free to be precise with decimals.

Ties are broken by order of submission

Prize: RedThunder K95 Mechanical Keyboard



The background is a solid blue color. On the left side, there are several overlapping circles of varying shades of blue, creating a decorative, abstract pattern.

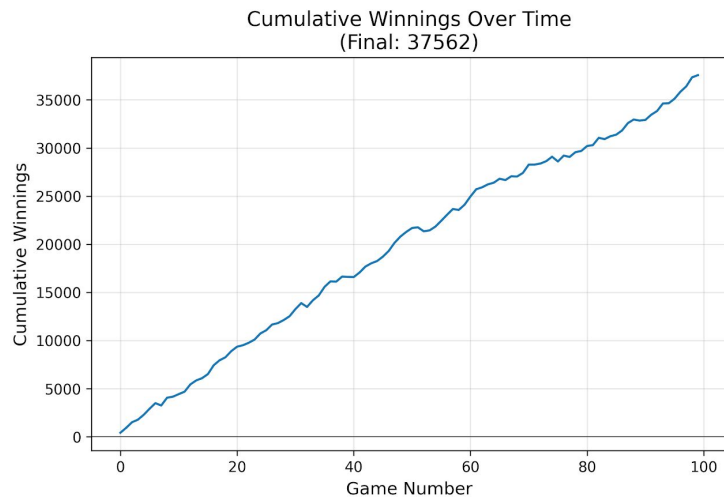
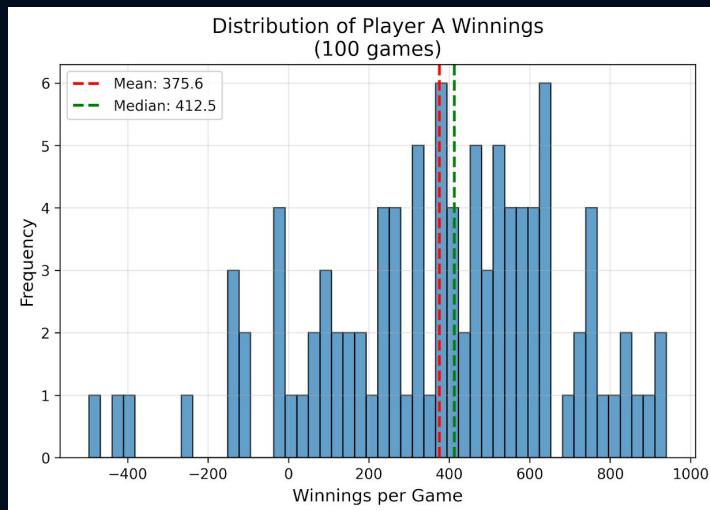
Announcements

Small Java Skeleton Bug Patched

- See pkr.bot/engine for most up-to-date engine-2026
- Keep letting us know about issues

Reference Bot Posted

- See pkr.bot/refbot1
- Tosses smallest card
- Betting identical to skeleton bots



Poker Social Tomorrow!

- Recitation block (location TBD, right after lecture)
 - Office hours will still run concurrently
- Great place to meet teammates!
- We will be giving away a ~surprise prize~
- Also try pkr.bot/teammates

Week 1 Deadlines

- First pokerbot submission due Friday 1/9, 11:59PM EST on scrimmage server
- Mini-tournament 1 will occur shortly after

Hackathon Next Week!

- Next Wednesday night (1/14)
- Show up and work on your pokerbot!
- Dinner provided, plan to go late into the night
- Snacks, fun, and games
- Prizes for challenges and those that stick around
- Location TBA

Outline of Today

- Probability Foundations
- Random Variables and Distributions
- Expected Value
- What Even is Statistics
- Law of Large Numbers
- Recitation Portion
 - Probability Practice Worksheet
 - Code-along reference-2 bot!
 - *Note that this portion will still be in 6-120, running from 1:30-2:30pm*



Probability Foundations

What is Probability?

- We say “probability of [event] happening is x ”
- x is usually a number from 0 to 1 or a percent
- But what does x 's value mean?

Interpretations of Probability

- Frequentist
 - How often an event occurs over repeated trials
 - Ex: When rolling a dice repeatedly, we notice 1/6th of rolls are a 2
 - What about events that aren't repeated? Eg: 2024 election
- Subjectivist (Bayesian)
 - Degree of belief, or 'credence' of an event occurring from the perspective of an individual with a given set of information
 - Different pieces of information 'update' the probability to be higher or lower depending on how much evidence they provide for the event
 - However, this requires some assumptions for which default probability values to start with before any updates
- This is mostly a philosophical question within epistemology
- Regardless of interpretation, probability is used as a quantitative metric for modeling uncertainty in things we don't know

Motivation: Finite Possibilities with Equal Chance

- Wish to find/define probability of some event that occurs in some outcomes
- An intuitive definition:
Probability of event = ($\#$ favorable outcomes) / ($\#$ total outcomes)
- Ex: Randomly draw card from standard deck, probability of Spades
 - 13 cards with spades
 - 52 cards total
 - $13/52 \rightarrow \frac{1}{4}$
- This reduces any question of probability into a counting problem
- What about outcomes with unequal chance?
- Or infinite outcomes? (e.g. continuous spectrum of possibilities)
- We need some general way to define their events and probabilities

Kolmogorov Axioms

Events are viewed as sets of outcomes, and every set is a subset of the largest set Ω , which can be viewed as the 'universal' set encompassing all outcomes. We define a function P to give the probability of an outcome falling within an event set, satisfying the following axioms:

1. $P(\Omega) = 1$
2. $P(E) \geq 0$ for any event set E
3. $P(A \cup B) = P(A) + P(B)$ for any disjoint (aka nonoverlapping) event sets A, B

Kolmogorov Axioms

These axioms can prove all other results which we'd expect to make sense:

- If $A \subseteq B$, then $P(A) \leq P(B)$
- $P(\text{empty set}) = 0$
- $P(E) \leq 1$
- $P(\Omega - E) = 1 - P(E)$

As long as we can define a universal set Ω and an idea of probability that follows the axioms, then we can use any of the results.

Example

- Universe of possibilities representing opponent's first-dealt card
- $P(\text{first card is } 2h) = P(\text{first card is } 2s) = \dots = 1/52$

Questions?

Conditional Probability

“probability of A given B”

$$P(A | B) = P(A \cap B) / P(B)$$

Another way to write: $P(A \cap B) = P(B) P(A | B)$

The basis of inference aka updating beliefs based on information.

Example (Toss or Hold'em)

- Universe of possibilities representing D (the event that our opponent tosses their lowest card) and H (the event that our opponent is left with two cards, each $\geq J$).
- We know they always toss their lowest card if they have at least two cards $\geq J$, and otherwise they toss their lowest with probability $\frac{1}{5}$.
- We want to find the conditional probability they are holding two cards $\geq J$ given that they tossed a 2 (must be their lowest).
- $P(D | H) = 1$
- $P(D | \neg H) = \frac{1}{5}$
- $P(H) = \text{unknown "c"}$
- Want to find $P(H | D)$

Example (Toss or Hold'em)

- $P(D \cap H) = ?$

D: opponent tosses a their lowest card

H: opponent is left with two cards, each $\geq J$

Example (Toss or Hold'em)

- $P(D \cap H) = P(D | H) * P(H)$

D: opponent tosses a their lowest card

H: opponent is left with two cards, each $\geq J$

Example (Toss or Hold'em)

- $P(D \cap H) = P(D | H) * c$

D: opponent tosses a their lowest card

H: opponent is left with two cards, each $\geq J$

Example (Toss or Hold'em)

- $P(D \cap H) = 1 * c$

D: opponent tosses a their lowest card
H: opponent is left with two cards, each $\geq J$

Example (Toss or Hold'em)

- $P(D \cap H) = c$

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Example (Toss or Hold'em)

- $P(D \cap H) = c$
- $P(D \cap \neg H) = ?$

D: opponent tosses a their lowest card
H: opponent is left with two cards, each $\geq J$

Example (Toss or Hold'em)

- $P(D \cap H) = c$
- $P(D \cap \neg H) = P(D \mid \neg H) * P(\neg H)$

D: opponent tosses a their lowest card

H: opponent is left with two cards, each $\geq J$

Example (Toss or Hold'em)

- $P(D \cap H) = c$
- $P(D \cap \neg H) = P(D \mid \neg H) * (1 - c)$

D: opponent tosses a their lowest card
H: opponent is left with two cards, each $\geq J$

Example (Toss or Hold'em)

- $P(D \cap H) = c$
- $P(D \cap \neg H) = \frac{1}{5} * (1 - c)$

D: opponent tosses a their lowest card

H: opponent is left with two cards, each $\geq J$

Example (Toss or Hold'em)

- $P(D \cap H) = c$
- $P(D \cap \neg H) = \frac{1}{5} - \frac{1}{5}c$

D: opponent tosses a their lowest card

H: opponent is left with two cards, each $\geq J$

Example (Toss or Hold'em)

- $P(D \cap H) = c$
- $P(D \cap \neg H) = \frac{1}{5} - \frac{1}{5}c$
- $P(D) = ?$

D: opponent tosses a their lowest card
H: opponent is left with two cards, each $\geq J$

Example (Toss or Hold'em)

- $P(D \cap H) = c$
- $P(D \cap \neg H) = \frac{1}{5} - \frac{1}{5}c$
- $P(D) = P(D \cap H) + P(D \cap \neg H)$

D: opponent tosses a their lowest card

H: opponent is left with two cards, each $\geq J$

Example (Toss or Hold'em)

- $P(D \cap H) = c$
- $P(D \cap \neg H) = \frac{1}{5} - \frac{1}{5}c$
- $P(D) = \frac{1}{5} + \frac{4}{5}c$
- $P(H | D) = ?$

D: opponent tosses a their lowest card
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- $P(H | D) = P(H \cap D) / P(D)$

D: opponent tosses a their lowest card
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D: opponent tosses a their lowest card
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Example (Toss or Hold'em)

- $P(D \cap H) = c$
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- $P(D) = \frac{1}{5} + \frac{4}{5}c$
- $P(H | D) = c / (\frac{1}{5} + \frac{4}{5}c)$

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Example (Toss or Hold'em)

- $P(D \cap H) = c$
- $P(D \cap \neg H) = \frac{1}{5} - \frac{1}{5} c$
- $P(D) = \frac{1}{5} + \frac{4}{5} c$
- $P(H | D) = c / (\frac{1}{5} + \frac{4}{5} c)$
 - i.e. $P(H) = c = 0.15$, $P(H | D) = 0.46875$

D: opponent tosses a their lowest card
H: opponent is left with two cards, each $\geq J$

Example (Toss or Hold'em)

- $P(H) = \text{low}$ (e.g. 0.15)
- $P(H | D) = \text{higher}$ (e.g. 0.46875)
- Takeaway: probability of an event increases when you observe evidence that makes it more likely
- Conditional probabilities allow us to mathematically represent this effect, which is called Bayes' Theorem

Questions?

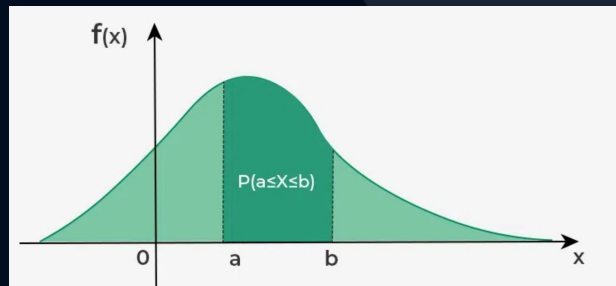
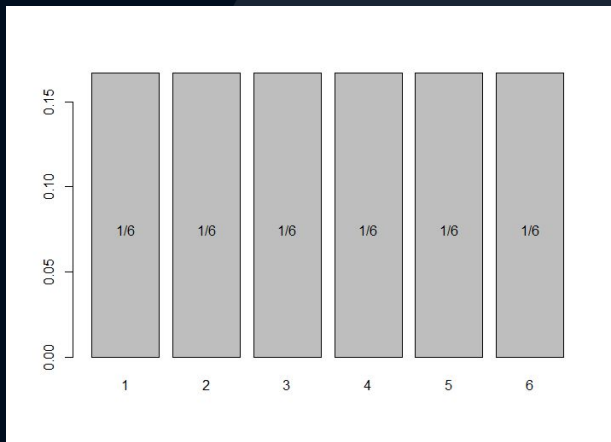
Random Variables and Distributions

Random Variable

- A random variable is a quantity or event which takes on different values with different probabilities
 - Ex: A drawn card that's face down and hasn't been turned over
- The set of possible values it can take on would represent the universal set of events for probabilities concerning this variable.
 - Ex: 52 total cards that our mystery card could have
- Any event concerning this variable corresponds to set possible values it takes on
 - Ex: "card is black" $\leftrightarrow$ set of 26 cards

Distributions

- Distributions are functions that are used to define probabilities for different events
- For discrete events, this is called a “probability mass function” (think bar graph)
- For continuous events, this is called a “probability density function” (think histogram)





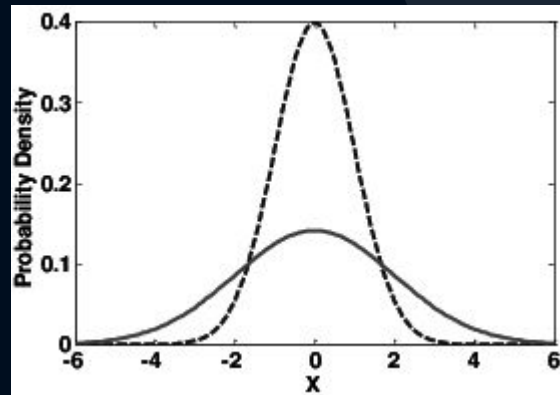
Expected Value

Expected Value

- The notion of expected value (EV) is used to summarize distributions on real numbers by some representative value
- In short, “expected value” is just another way to say “mean”
- Finite events with equal probability → take the average
 - EV of dice roll: $(1+2+3+4+5+6)/6 = 3.5$
- What about events with different probabilities?
- Generalize to weighted average (weighted by probability)
 - X is a random variable with possible values in set S
 - $E[X] = \sum_{s \in S} s * P(X = s)$
- For continuous distributions, this generalizes to an integral calculation

Variance

- If expected value describes where random variable “usually” is, then variance measures how much the random variable may fluctuate
- This is done by finding the average squared distance from the mean:
 - Random variable X with $E[X] = \mu$
 - $\rightarrow \text{Var}[X] = E[(X - \mu)^2]$
- Standard deviation is a similar measurement, which is simply the square root of variance



In short

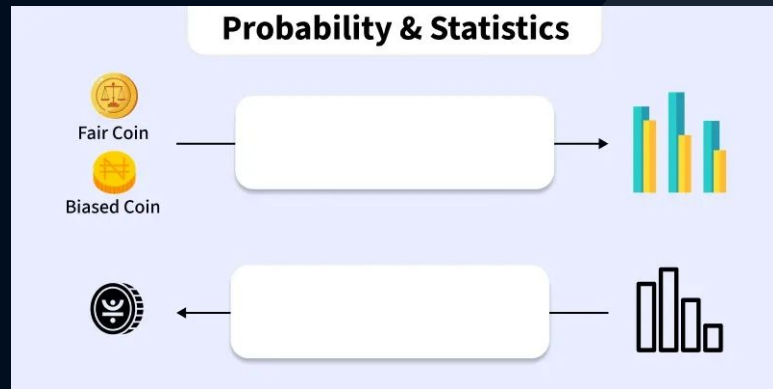
- Expected Value is the average of a random variable and is often used as a “best guess” for the result
- Variance and standard deviation are used to measure how close this guess would generally be



What Even is Statistics

A Bridge Between Two Worlds

- Where does statistics come into the picture?
- Probability and Statistics are dual
- Oftentimes we want to predict the future given the past
- Possible approach:
 - Use statistics to infer the data-generating distribution from the past
 - Use probability to predict the data of the future



Law of Large Numbers

Law of Large Numbers

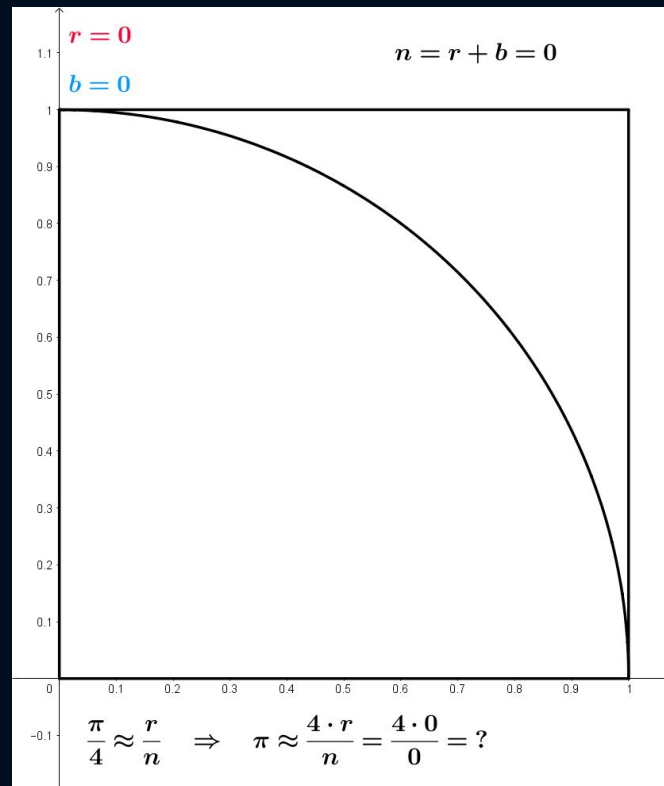
- X is a random variable with mean μ with some distribution
- Independent samples $X_1, X_2, \dots, X_n$ are drawn from the distribution.
- “Sample mean” $\bar{X}$ is the average of these samples.
- It can be shown that $E[\bar{X}] = \mu$ and $\text{Var}[\bar{X}] = \text{Var}[X]/n$
- The law of large numbers states that this sample mean $\bar{X}$ is guaranteed to approach the actual mean μ as n (the number of samples) approaches infinity

Law of Large Numbers

- This gives us another intuitive way to think of the mean - the average of infinite hypothetical trials
- The law of large numbers also conveys a powerful idea: with enough data points, you can accurately estimate properties of random processes, even if their underlying distributions are unknown
- With the ability of computation, certain quantities are now much easier to compute

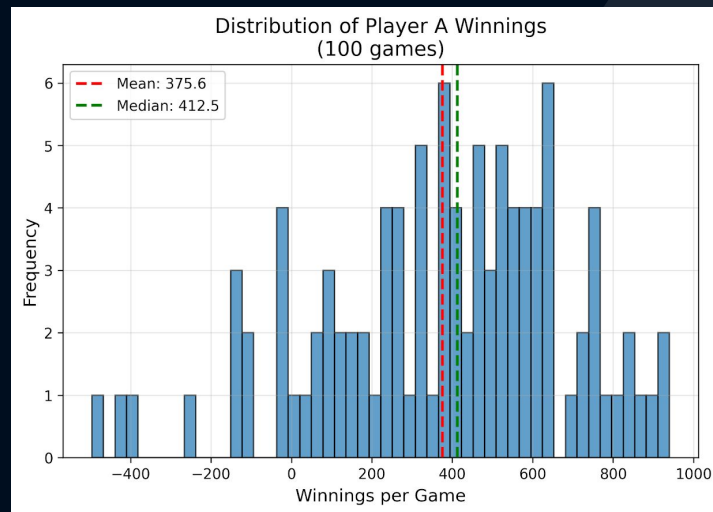
Monte Carlo Estimation

- Style of computational methods to calculate a result using repeated random sampling and averaging (aka a direct application of LLN)
- Ex: estimating area using geometric probability



Monte Carlo Estimation

- Another example is the way we run Pokerbots matches!
- Dividing the cumulative deltas by 1000 is a Monte Carlo estimate for the expected number of chips gained in a single round
- Therefore a good pokerbot should try to maximize the expected number of chips earned within a single hand





Summary

Summary

- Probability is a tool for describing uncertain situations
- In particular, Expected Value gives us a numerical estimate of an average scenario
- Statistics allows us to estimate distributions given data drawn from them
- Law of Large Numbers and Monte Carlo use generated samples to make some calculations easy with computation
- If you'd like to learn how to use numerical/statistical packages to do some of these computations, come to recitation on Thursday!

Lunch Time!

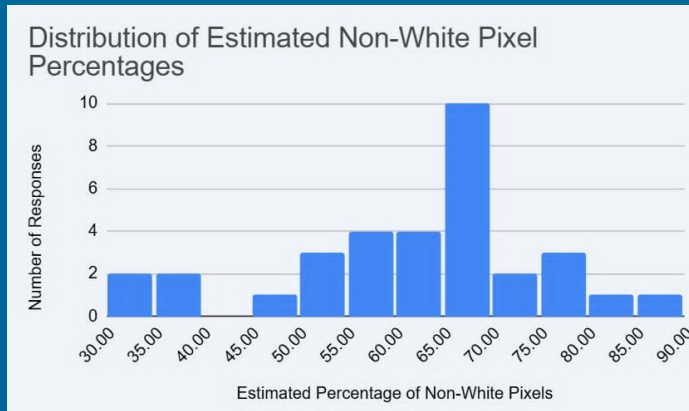
Leave any type of feedback at pkr.bot/feedback!





Coding Demo

Clubs Game results



Answer: 54.2550%

Congrats to:

shagun12

(guess of 53.33)

Fun fact

most common
answer:

67.67 (2 instances)

Thanks for watching!

Slides will be posted on pkr.bot/resources

Make sure to check pkr.bot/piazza for updates

Lecture recordings at pkr.bot/panopto

Leave feedback at pkr.bot/feedback!



Recitation!